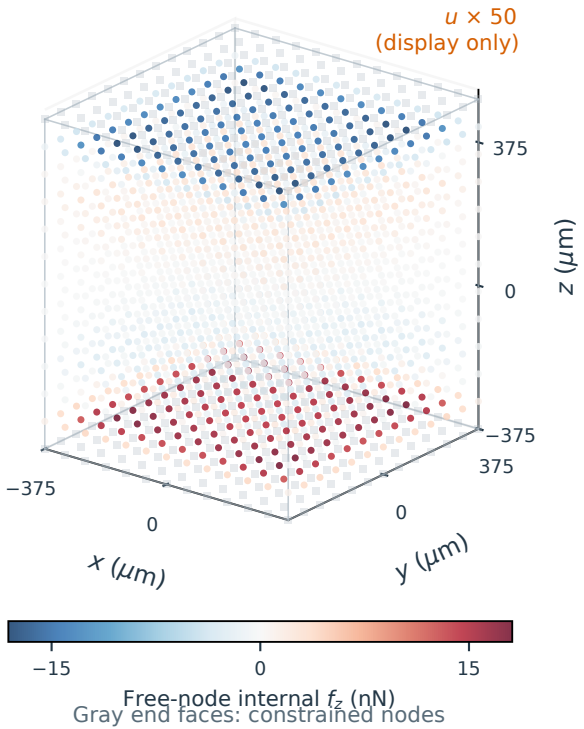
3. Trace the Force Imbalance Near Edges

Existing L075 affine state | Strain 0.3% | 2,197 particles | Phase disabled

(a) Uniform stretch, nonzero free-node force

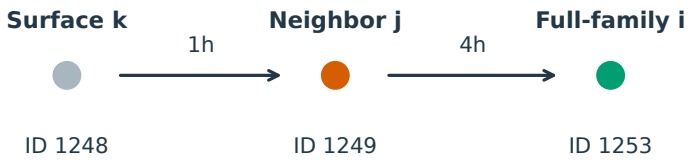


Gray end faces: constrained nodes

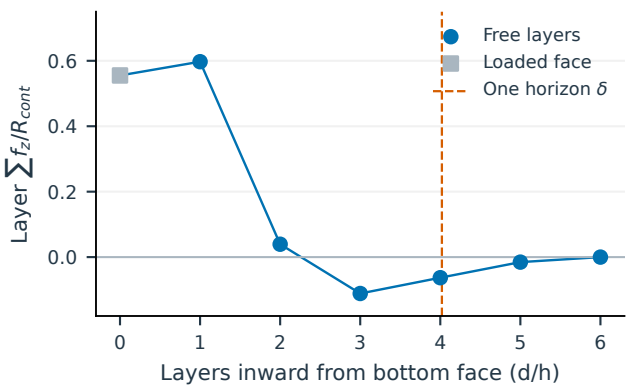
Max free-node force: 18.11 nN

Accurate deformation is not force equilibrium.

(b) Neighbor dependence reaches inward



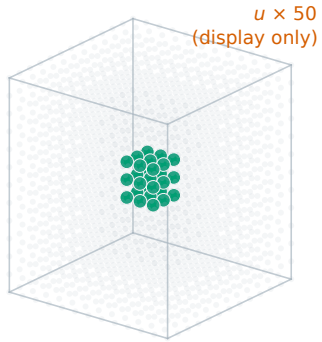
One real dependency path; layout not to scale



Selected core particle 1253:

Neighbor Z term: 95.9% of signed f_z

(c) Boundary effect or bulk defect?



27 geometrically full families

0 closed two-hop / three-hop cores

Core max |f| / R = 1.07e-04

Noise criterion: 1e-10 → FAIL

Boundary influence or bulk defect?

Physical cause: UNRESOLVED

Static internal-force evidence. Dependency arrows do not show a time-evolving propagation front or per-link force attribution.

Constraint reactions must be distinguished from unbalanced free-node forces. Geometric full families do not isolate the complete force stencil.